

The Italian Stagnation: A Multi-Lens Visual Investigation

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1 Introduction and question

The Italian wage anomaly is a familiar fact in European policy debate but rarely a familiar shape. Across Western Europe, real wages have grown over the last three decades by anywhere from roughly five to more than sixty percent. In Italy, the figure is barely three. **Why has Italian real wage growth stalled, and what does that cost individual workers?**

This project answers the question through a six-figure interactive newspaper feature, built for an educated non-specialist audience and designed to be read linearly as a story or used for exploration. The investigation approaches wages through six complementary analytical lenses (country, region, firm size, generation, taxation, and personal experience), each implemented as a dedicated visualization. The taxation component is a deliberately paired *white-hat* / *black-hat* treatment of the same dataset, so that the project closes by interrogating its own visual choices.

2 Data

The investigation rests on seven public datasets, each of which was retrieved by hand from the OECD, Eurostat and ISTAT browser portals with explicit filter selections, plus a small hand-curated consumption-price list compiled for the personal calculator. For every dataset we ran an exploratory data analysis (EDA) before committing to a visualization choice, both to understand its coverage, granularity and shape, and in several cases to discard the original chart hypothesis when the data did not support it (the resulting alternatives are logged in §5).

T1. OECD Average Annual Wages. Real annual earnings per full-time-equivalent worker, in constant 2024 USD at purchasing-power parity, 1990 to 2024, covering Italy and four Western European peers (Germany, France, Spain, Sweden) plus the OECD aggregate as a reference baseline. Provides T1's headline country-level trajectory comparison. Chosen because it is the only publicly available wage indicator in which the same PPP-comparable real-wage methodology is applied consistently across countries; without that consistency, the cross-country chart would be comparing apples to pears.

T2. Eurostat regional and country-level GDP per capita. Two companion indicators from Eurostat's national-accounts family. *nama_10r_2gdp* gives GDP per capita in PPS at the NUTS-2 level, 2000–2024, for the seventy regions covering Italy, Germany, France and Spain. This is T2's choropleth backbone. *nama_10_pc* gives the Eurozone (EA20) country-level series used as the colour-scale threshold. Switching the reference from EU27 to EA20 was a deliberate choice for internal consistency with the Western European peer scope of the rest of the project (discussed in §3). The map's geographic boundaries are taken from Eurostat's NUTS-2 cartographic file.

T3. Eurostat structural business statistics. Indicator *sbs_sc_sca_r2*: employment shares and labour productivity by firm size class (1–9, 10–19, 20–49, 50–249, 250+), four economies (Italy, Germany, France, Spain), 2020 snapshot. The joint distribution of firm size and productivity is the empirical core of T3's test of the *nanismo* hypothesis: whether Italian firms are unusually small and unusually unproductive within each size class.

T4. ISTAT RACLI and Eurostat HICP. **ISTAT RACLI** (*Rilevazione su Costo Lavoro e Retribuzioni*) is the only national-source dataset in the project: median gross hourly wage of Italian dependent workers by age band (15–29, 30–49, 50+), 2014 to 2023. RACLI is the only Italian public wage source that disaggregates by age consistently across this window and that includes micro-firms (which Eurostat's earn-ses excludes), an inclusion that is non-negotiable for the Italian case. The Eurostat HICP series *prc_hicp_aind* is used as the price-level deflator that converts each annual nominal RACLI value into constant 2023 euros, so that within-country wage decline cannot be mistaken for inflation.

T5. OECD Wages (extended set) and concrete-unit prices. The calculator uses the same OECD Average Annual Wages source as T1, extended to a wider Western European set: Italy plus seventeen other European OECD members for which a continuous, methodologically consistent real-wage series is available across the relevant span. To translate the cumulative monetary gap into something tangible, we curated a small 2024-euro price list of consumption units (Fiat Panda, monthly Milan rent, weekly Mediterranean holiday, Vespa, pizza *margherita*); the individual sources are listed inline in the calculator’s data file.

T6. OECD Taxing Wages. Personal income tax, employer and employee social-security contributions, and the resulting tax wedge for a range of household types and income brackets, eighteen European OECD countries, 2024 snapshot. Used to build the white-hat / black-hat pair on the politically charged “tax wedge” concept. The very flexibility of the dataset (multiple denominators, multiple income levels) is itself the editorial subject of T6.

Integration. Every chart consumes one or two harmonised input panels. T1 indexes every country to its 1995 value to strip out level differences and isolate growth dynamics. T2 keeps a hand-picked whitelist of seventy NUTS codes (Italian *regioni*, German *Länder*, French post-2016 *régions*, Spanish *comunidades*) plus EA20 as reference, discarding redundant macro sub-aggregates and overseas territories that would distort the European picture. T4 deflates each annual ISTAT median wage by the corresponding HICP ratio against 2023, yielding constant-2023-euro values directly comparable across the decade. T5 inlines the cleaned wage indices and the consumption-unit price list into the JavaScript bundle so the calculator runs offline as well as online.

Limitations. Five matter for the analysis. (i) *Coverage windows differ across datasets:* T1 begins in 1995, T2 in 2000, T4 in 2014, T3 and T6 are single-year snapshots, so claims that span the full period are framed explicitly per chart rather than asserted globally. (ii) *ISTAT publishes only three coarse age bands;* a finer five-year grid exists in INPS reports but only as scanned annual PDFs without bulk download. (iii) *The OECD wage series is reported in USD PPP rather than euro,* so trajectories are comparable across countries but absolute levels are not directly readable as Italian salaries (T5 partially addresses this by accepting an Italian-euro override, defaulted to the ISTAT median). (iv) *T3 covers a single year (2020):* the *sbs_sc_sca_r2* indicator’s most recent fully-populated cross-country cut at the time of writing; the firm-size structure is treated as slow-moving and we accept this as a snapshot. (v) *The tax-wedge value in T6 is highly sensitive to the chosen income bracket and denominator;* precisely the vulnerability the white-hat / black-hat pair is designed to demonstrate.

3 Audience, tasks, and design rationale

Audience. The system is built for an educated, non-specialist reader of long-form economic journalism: the audience of *Il Post*, *Domani*, *Il Sole 24 Ore* and similar Italian and European outlets. This reader follows headline economic numbers without working with the underlying datasets. The system therefore privileges narrative clarity over analytical control: every chart is captioned and introduced by editorial prose, and interactivity is offered as a complement to the reading experience, never as a substitute for it.

Tasks. The article supports six analytical tasks, one per visualization. Each task is anchored to an explicit research question, listed in italics below:

- **T1. Country trajectory.** *Has Italy fallen behind its Western European peers in real wages over the last three decades?* The chart reads Italy’s real-wage path against four peers and the OECD aggregate.
- **T2. Regional decomposition.** *Is the stagnation confined to the Italian Mezzogiorno, or have Northern regions also fallen behind their European peers?* The chart maps NUTS-2 regional growth against the Eurozone average.
- **T3. Structural decomposition.** *Italian firms are unusually small. Is the productivity problem the smallness, the underperformance of small firms, or both?* The chart shows the joint firm-size and productivity distribution across four economies.
- **T4. Generational decomposition.** *Has the wage stagnation hit some age bands harder than others?* The chart tracks real hourly wages by age band across the post-2014 decade.
- **T5. Personal cumulative loss.** *What is the cumulative personal cost of the Italian wage stagnation, for a given start year and country comparator?* The calculator converts the aggregate gap into a user-specified loss, expressed in concrete consumption units.
- **T6. Editorial sensitivity.** *Can the same OECD tax-wedge dataset support opposite editorial readings depending on a few editorial choices?* The white-hat / black-hat pair compares two presentations of the same dataset to surface the four levers that turn the same numbers into opposite stories.

A cross-chart link binds T1 to T5: clicking a country line in T1 auto-fills the comparator in the personal calculator, so a structural finding (“Italy is sixty index points behind Sweden”) flows directly into a personal one (“this would have

been worth X to me”).

Design rationale. We built the system as an article one reads, not as a dashboard one queries. Every figure shares the same typography and the same warm off-white background, so that the six charts feel like they belong to a single newspaper page rather than six separate apps. Within that shared frame, each chart’s form was chosen for what it lets the reader see immediately.

T1 is the only line chart in the system, justified by its thirty-year temporal axis: a multi-country indexed line chart is the form that most directly answers “when did the divergence begin?” T2 is a choropleth, the natural form for cross-regional comparison; the EA20 average is encoded as the colour-scale midpoint so that red/green reads as “below/above European average” at a glance. T3 is a Coxcomb (Florence Nightingale) polar chart, an unconventional choice that compresses two dimensions of the *nanismo* hypothesis (firm-size share as radial thickness, productivity as colour) into one frame in a way two side-by-side standard charts cannot. T4 is the project’s “creative” visualization: a stacked-coin pictogram in which one coin is one euro of real hourly wage, so that a fifteen-percent decline appears as two coins removed from a stack, an emotionally legible move that a column chart cannot make. T5 is the required JavaScript component, an interactive D3.js calculator that takes the reader’s input and outputs a personal loss in concrete units. T6 is a paired bar chart on purpose: the bar form is rhetorically persuasive for ranking comparisons, which is precisely why the white-hat / black-hat exercise stages it.

4 System description

The system is published as a single scrollable webpage on GitHub Pages, organised as a six-figure narrative. The figures appear in the order the reader encounters them, which differs slightly from the technical task numbering: T1, T2, T3, T4, then T6 (the white-hat / black-hat pair, used as a fifth analytical lens on taxation), and finally T5 (the personal calculator, used as a closing). Each figure is wrapped by editorial prose: a setup paragraph that asks the research question, the figure itself, then a findings paragraph that interprets what the reader has just seen.

T1. Indexed line chart (Plotly). Real-wage trajectories for Italy, four peers (Germany, France, Spain, Sweden) and the OECD aggregate, indexed to 1995. Three preset buttons (*Full period*, *Pre-2008*, *Post-2008*) switch the time window and refresh annotations. Hovering any line reveals a country-year-value tooltip; clicking a non-Italy line dispatches a `postMessage` that drives the cross-link to T5 (described below).

T2. Choropleth (Tableau Public). Seventy NUTS-2 regions of Italy, Germany, France and Spain, colour-coded by their 2000–2024 GDP-per-capita growth. The colour scale is centred on the Eurozone (EA20) growth rate, so red and green read directly as “below/above European average”. Hover reveals the region name and the exact growth rate. The choropleth is hosted on Tableau Public and embedded into the page through a Tableau viz object.

T3. Coxcomb / Florence Nightingale rose (Plotly). Four countries occupy quadrants of a polar plot; within each quadrant, five concentric size-class rings stack outward from *micro* (0–9 employees) at the centre to *very large* (250+) at the rim. Radial thickness encodes the country’s employment share in that size class; colour encodes labour productivity on a 20–80 k€/worker scale. Two interactions are implemented: a row of HTML checkboxes above the chart filters the four countries on/off and redistributes the remaining quadrants evenly around the circle, while clicking any country expands it to fill the full circle as a single five-ring donut.

T4. Animated pictogram (Plotly). Stacked-coin frames in which one coin is one euro of real hourly wage, for three featured years (2014, 2021, 2023) and three age bands (15–29, 30–49, 50+). A “► Watch the decade” button animates the year-by-year reveal so the post-2021 cliff is staged narratively; a dropdown filters to a single age band.

T6. White-hat / black-hat static pair (matplotlib). Two static PNGs of the OECD tax-wedge data. The white-hat figure shows the canonical OECD methodology: tax wedge as a percentage of total labour cost, all eighteen European OECD countries, sorted ascending, European median marked. The black-hat figure shows the same dataset with four levers flipped (cherry-picked twelve-country selection, 167% of average wage instead of 100%, gross-wage denominator substituted for labour cost, populist title). The pair is static by design: the rhetorical move is in the data choices, not in any interactive feature.

T5. Personal calculator (D3.js, the JavaScript component). The required original JavaScript artefact. The widget takes three user inputs (start year via slider, comparator country chosen from a dropdown of eighteen European OECD members, optional override salary in euros) and computes the year-by-year cumulative real-wage gap between the user’s Italian trajectory and the comparator’s, integrated into a single euro figure. The output is rendered three ways: as the

single cumulative number; as a year-by-year line chart of implied annual real salary (drawn with D3.js into an inline SVG); and as the same loss translated into concrete consumption units (Fiat Pandas, months of Milan rent, weekly Mediterranean holidays, Vespas, pizzas) chosen via a dropdown.

Cross-chart link. The brief requires that at least four of the visualizations be “interactive and linked”; the link in this project binds T1 to T5. When the reader clicks a country line in T1 (which lives in its own iframe), the embedded chart posts `{event: 't5:set_country', iso3: ...}` to the parent page; the parent forwards the message to T5’s iframe, T5 updates its country dropdown and re-runs the calculation, and the page smooth-scrolls to the calculator. The two iframes never talk directly: the parent acts as a relay, which keeps both charts self-contained.

Visual identity. All six figures share a single visual system: a warm off-white background (#f8f5ec), Playfair Display for headlines, EB Garamond for body, and an Italian-red / neutral-grey / forest-green palette in which red consistently marks Italy throughout. Interactive conventions are also shared: hover for tooltips, click for drill-down or cross-chart action.

5 Rejected alternatives

Three alternatives are worth recording because rejecting each one made the project change shape.

1. EU27 as the European reference, project-wide. The default European aggregate that statistical agencies expose to data journalists is EU27 (the twenty-seven current EU member states), and most published charts that compare an Italian indicator to “the European average” use it. We considered using EU27 throughout the project as our peer benchmark and rejected it. Over the 2000–2024 window, EU27 GDP per capita in PPS grew by roughly 117 %; the EA20 (Eurozone) average grew by 98.6 %; the difference, about twenty index points, comes almost entirely from the post-2004 accession countries (Poland, Czech Republic, Romania, Bulgaria, Hungary, the Baltic states), which had catch-up growth of +150 to +250 % from a much lower 2000 baseline. Comparing Italy to that average is comparing a mature high-income economy to a continent in the middle of a one-time convergence: it would have made every Italian indicator look worse than it really is against similar-development peers, and it would have contradicted T1’s selection of nine Western European OECD members. We chose Western European peers and the EA20 aggregate as the project’s consistent benchmark, and EU27 is mentioned only where its absence would be conspicuous (in the T2 caption, for example).

2. Three rejected chart types for T3. The firm-size lens went through three abandoned chart types before we settled on the Coxcomb. The first attempt was a hierarchical *sunburst* (country at the centre, size class at the rim), which told the “Italy is mostly micro-firm” story but compressed productivity to a single uncalibrated colour, hiding the second half of the *nanismo* hypothesis (that Italian micro-firms also produce less than peer micro-firms). The second was a *bubble scatter* of size class on the x-axis against productivity on the y-axis, with bubble area encoding employment share: it carried both dimensions but produced a chart with no narrative anchor, no reason for the eye to stop on Italy. The third was a productivity *heatmap*: clean for one dimension but lost the employment-share story entirely. The Coxcomb (Florence Nightingale polar chart) is the form we kept because it encodes share radially, productivity by colour, and country by quadrant in a single image, so the joint fingerprint of Italian *nanismo* reads at a glance.

3. Alternative wage sources for T4. ISTAT RACLI was not our first attempt at the generational lens. We tested three other public sources before settling on it. Eurostat *earn_ses_pub2a* (Structure of Earnings Survey) covers 2014–2018–2022 with finer breakdowns but excludes firms with fewer than ten employees by design, a non-trivial omission for Italy, where micro-firms employ 42 % of the workforce and earn substantially less than larger firms. The INPS *Osservatorio Lavoratori Dipendenti* publishes five-year age bands (the granularity we wanted) but only as eleven separate annual PDF reports without a bulk download. The Bank of Italy’s SHIW (*Indagine sui Bilanci delle Famiglie*) would have allowed birth-cohort tracking back to 1977, but its tables are equivalised at the household level rather than the individual, and its individual microdata require an academic application. RACLI’s simpler three-band structure (15–29, 30–49, 50+) is coarser than ideal, but it is the only Italian source that includes micro-firms, covers 2014 to 2023 consistently, and is published as a downloadable cross-tab.

6 Findings, limitations, and the white-hat / black-hat reflection

The question, retrieved. The project addresses one of the most controversial and most-discussed economic facts of contemporary Italy: that Italian wages, the public debate insists, have not really moved in a generation. The conversation around the claim is loud, often hyperbolic, and almost never comparative. “*In Italia gli stipendi sono fermi*” is a sentence that, in Italian discourse, needs no follow-up. The question we set out to answer is the follow-up: are they really? And if so, what does the rest of Western Europe look like in the same window?

Findings. The answer is yes, with caveats, and the caveats are the project’s contribution. Italian real wages have not in fact been still: they grew by roughly 3 % over 1995–2024. But that 3 % is the only growth figure in the room: every Western European peer the project measures has grown faster, and most of them have grown far faster (Spain +5 %, Germany +22 %, France +27 %, the OECD average +31 %, Sweden +62 %). The Italian line peaks around 2010 and then falls back to roughly its 1995 level by 2024, the only major Western European economy to do so.

When that aggregate is broken down geographically, the popular Italian explanation, that it is the *Mezzogiorno* that drags the country down, collapses on the regional map. Twenty of Italy’s twenty-one regions grew more slowly than the Eurozone (EA20) average over 2000–2024, and the slowest is not in the South but in the centre of the country. Lombardia, the Italian economic engine, grew twenty-six percentage points below the Eurozone average. The structural answer, made visible by the firm-size lens, is double-edged: Italian micro-firms employ twice the share of workers German or French micro-firms do (42 % against 19 % and 23 %) and they produce roughly 60 % of what their German peers produce per worker. Italy is not merely unusually concentrated in low-productivity firms; those firms are themselves unusually low-productivity by European standards.

The generational lens completes the geometry. Between 2014 and 2023, every Italian age band loses real-wage ground (between 9 and 13 percent), with the senior band losing most, contrary to the popular intuition that tenure shelters older workers. The decline is concentrated in 2021–2023, the years in which fifteen percentage points of cumulative inflation went largely unabsorbed by collective bargaining. Translated through the calculator into a personal cumulative loss, those aggregates resolve into a five-figure number for any given reader: enough to convert into multiple Fiat Pandas, years of Milan rent, or thousands of pizza *margherite*. The aggregate fact has a face.

The white-hat / black-hat reflection. The taxation lens (T6) was built as a deliberately reflexive figure. The white-hat chart shows the canonical OECD tax wedge for a single worker at 100 % of average wage: Italy at 47.0 %, fifth of eighteen, clustered with Belgium (52.4), Germany (47.9), France (47.1) and Austria (47.1), the European welfare-state group. The black-hat figure takes the same OECD dataset and applies four levers: cherry-picked comparators (twelve countries, none that would beat Italy on the manipulated metric); income level switched from 100 % to 167 % of average wage; denominator switched from total labour cost to gross wage; and a populist title (“Italy’s true tax burden on workers”), where “true” is the rhetorical hinge. The Italian number jumps from 47 % to 71 %, and the visual reading shifts from “high but in line with peers” to “anomalous and abusive”. Both figures are mathematically defensible; only one is honest. The lesson the pair stages, for our own project as much as for its audience, is that the same data does whatever its author wants, and that the guardrails we have tried to maintain throughout (canonical denominators, consistent peer sets, transparent captions) are the same guardrails the black-hat figure violates, one by one.

Limitations and what this paper does not claim. The most important limitation is the easiest to lose sight of when the charts look clean. The project documents *correlations*: Italian wages are flat, regional growth is uniformly low, micro-firms dominate Italian employment, the decline is generationally uniform, the tax wedge is high in European context. It does not document *causes*. The explanations that fill the public Italian debate on this question (the geopolitical reordering of Europe after the 1990s, the structural and administrative fragility of the Italian state, the country’s chronic political instability, the impact of the 2020 pandemic, the energy shock of 2022, demographic ageing) are absent from this analysis. They are not absent from the actual story; they are absent from *ours*. The artefact is therefore offered as a journalistic investigation, not as a causal model: it documents the shape of the Italian wage anomaly with care, but it does not, and could not honestly, claim to explain why that shape exists.

The data itself imposes additional boundaries. The project uses aggregate medians and shares, so it cannot speak to the underlying distribution: the same headline number can hide widening inequality or contract-type substitution that only microdata would surface. The timeline is a mosaic (1995, 2000, 2014, 2020, 2024) that does not support fully synchronous causal claims across the six lenses. The peer set is restricted to Western European OECD members, whereas extending to the United States, Japan or South Korea would help distinguish a European phenomenon from a Western-developed one. Future improvements would weave each of these missing dimensions into a richer story, ideally with microdata access for the personal lens and birth-cohort tracking for the generational one.